

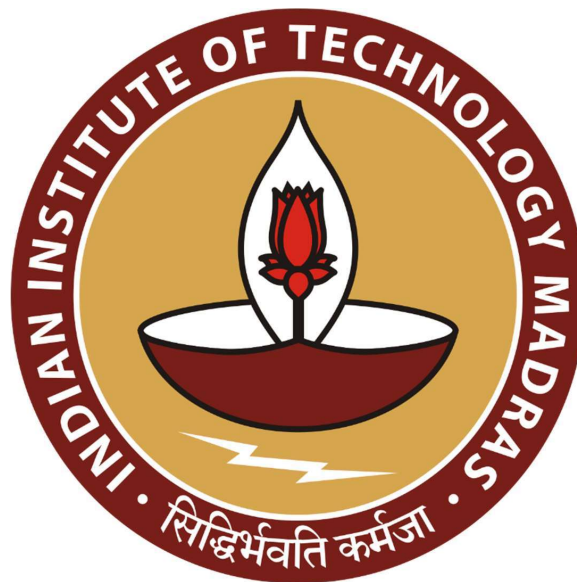
A Business Data Management Report on Sales Insights and Operational Challenges in a Local Bangle and Wool Store

A Final Term report for the BDM capstone Project

Submitted by

Name: Vivek Mittal

Roll number: 23F1001475



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents

1. Executive Summary
2. Detailed Explanation of Analysis Process/Method
3. Results and Findings
4. Interpretation of Results and Recommendation

1. Executive Summary

Project Title: “A Business Data Management Report on Sales Insights and Operational Challenges in a Local Bangle and Wool Store”

This project presents a detailed analysis of the sales performance and operational patterns of Naina Kangan and Wool Store, located at New Market, Doiwala, Dehradun (248140), Uttarakhand, a small retail and wholesale business dealing primarily in bangles and wool-related products. The study was undertaken to understand existing sales behavior, identify key operational challenges, and explore how data-driven analysis can support better decision-making in a traditionally managed business setup.

The analysis is based on primary sales records maintained by the business owner in handwritten registers. These records were digitized into structured Microsoft Excel sheets and analyzed in two stages across the mid-term and final-term phases. The final-term report builds upon the validated mid-term dataset without introducing assumptions or artificial adjustments. During final verification, 46 zero-revenue closed-day rows were separated from the sales records and one mis-tagged entry (a wool ball sale recorded under bangles) was corrected; no sales values were altered.

The dataset covers a total revenue of ₹12,19,396 generated through 2,378 sales transactions between 7 February 2024 and 31 January 2025, involving both wholesale (B2B) and retail (B2C) customers. Descriptive statistical methods, time-based analysis, product-type-wise evaluation, customer segmentation and ABC-XYZ inventory classification were used to study sales trends, demand patterns and revenue concentration. Bar charts, line graphs, a histogram and a Pareto curve support the findings.

The results show a highly seasonal business. Revenue peaks in the festive and monsoon periods: October (₹3,19,897) and July (₹2,09,570) together account for 43% of annual revenue, while September is the lowest month at ₹24,990. Winter generates ₹2,93,366 (24% of revenue) and is driven by wool, which makes up 72% of winter revenue. Bangles account for ₹8,97,620 (73.6%) of total revenue and carry the higher gross margin (24.2% against 15.1% for wool), largely through wholesale transactions.

The study also identifies operational inefficiencies arising from unsystematic stock handling and heavy reliance on cash, which accounts for about 95% of transactions (96% of revenue). While this ensures immediate liquidity, it restricts financial traceability and increases dependence on manual records. Revenue is concentrated more by product and season than by individual transaction: Bangles Box alone contributes 50.7% of revenue, while the top 20% of transactions contribute 44.6%. The ABC-XYZ analysis places four products in Class A (79% of revenue) and finds that no product has stable demand.

Based on these findings, the report recommends simple, practical measures: seasonal demand forecasting, stocking by ABC-XYZ class, basic customer record-keeping, and gradual adoption of digital payments. Overall, the analysis demonstrates that even basic data-driven practices can help the business improve efficiency, manage seasonal risk, and strengthen long-term sustainability.

2. Detailed Explanation of Analysis Process/Method

The analysis presented in this report is based on primary sales data obtained directly from the operational records of Naina Kangan and Wool Store. These records were originally maintained manually by the business owner in handwritten registers. During the initial phase of the project, they

were systematically digitized into structured Microsoft Excel sheets through careful transcription, verification and formatting, so that the original transaction details were accurately preserved.

For the final-term analysis, emphasis was placed on using the already cleaned and validated dataset developed during the mid-term phase. This ensured continuity across stages and allowed the final analysis to build directly upon previously verified data. No artificial adjustments, assumptions or synthetic values were introduced. Maintaining data authenticity was a key priority so that the findings reflect real business behavior rather than modeled outcomes.

Nature and Scope of the Dataset

The dataset consists of transaction-level sales records covering both wholesale (B2B) and retail (B2C) customers. Individual customer identifiers were not available in the original records, which mirrors the operational reality of the business. Transactions are recorded with product type, quantity sold, selling price, cost price and payment mode. This limits customer-level tracking but is sufficient for understanding overall sales patterns, revenue behavior and operational trends.

The source sheet contains 2,424 rows. Of these, 46 are days on which the store recorded no sales (holidays or closures). These days were retained in a separate list to preserve the continuity of the calendar and an accurate view of low-demand periods, but they are not counted as transactions. All transaction counts and per-transaction averages in this report therefore use the **2,378 sales transactions**. Two entries with inconsistent product-name capitalization were standardized, and one wool ball sale recorded under the bangles category was re-assigned to wool.

Variable Classification with Analytical Framework

To improve analytical clarity, the dataset variables were grouped into four dimensions:

- Time-related variables: date, month and season (Winter = November to February, Spring = March–April, Summer = May–June, Monsoon = July–September, Autumn/Festive = October)
- Product-related variables: product category and product type
- Financial variables: selling price, cost price, total revenue and gross profit
- Transaction-related variables: customer type (B2B/B2C) and payment mode (cash/online)

Customer type is as recorded by the owner: Bangles Box, Bangles Set, Fancy Bangles Box and Metal Bangles sales are recorded as B2B (1,308 transactions), while Bangles Box (Plus size) and all wool and accessory sales are recorded as B2C (1,070 transactions).

Descriptive Statistical Analysis

Descriptive statistical analysis formed the foundation of the final-term study. Key numerical variables such as quantity sold, total revenue and gross profit were summarized using measures of central tendency to establish baseline performance across products and customer types. Measures of dispersion such as minimum, maximum and variation were used to capture the spread of transaction values, which is important given the mix of frequent low-value retail transactions and infrequent high-value wholesale purchases.

Frequency and Percentage-Based Analysis

Frequency and percentage analysis was applied to categorical variables including product category, customer type, season and payment mode. This distinguishes transaction volume from transaction value: wholesale customers generated more transactions (1,308 against 1,070) and also a disproportionately large share of revenue (68%) because of their higher value per transaction.

Time-Based and Seasonal Analysis

Given the strong seasonal dependence of the business, time-based analysis was conducted at monthly and seasonal levels. Total quantity sold, total revenue and gross profit were calculated for each period to identify sales concentration and demand cycles, and a monthly seasonal index (month revenue divided by the average month) was computed to support purchase planning.

Segmentation by Product and Customer Type

The dataset was segmented by product category and customer type to compare sales volume, transaction frequency and profitability. Wholesale customers contribute most of the revenue through bulk purchases, while retail customers contribute smaller, steadier transactions; analyzing them together highlights the mixed nature of the business.

Inventory Classification: ABC-XYZ Analysis

Product types were classified by revenue contribution (ABC: Class A up to 80% of cumulative revenue, Class B up to 95%, Class C the remainder) and by demand variability (XYZ: coefficient of variation of monthly units sold, $X \leq 0.5$, $Y \leq 1.0$, Z above 1.0). The combination identifies which products deserve priority stock and which carry the most demand uncertainty. RFM analysis could not be performed because customer identifiers were not recorded; basic customer record-keeping is recommended to enable it in future.

Analytical Tools and Software

Excel was used as the primary analytical tool throughout the project: standard statistical functions, aggregation formulas, pivot tables and charting. Excel was chosen for its suitability for manually digitized datasets and the transparency of its calculation logic, which allows all computations to be verified and replicated. The headline figures were additionally reproduced independently in Python to confirm accuracy.

3. Results and Findings

This section presents the key results derived from the analysis of transaction-level sales data collected from Naina Kangan and Wool Store. The findings are supported by graphical representations and focus on trends, patterns and structural characteristics related to sales performance, product mix, seasonality, customer behavior and revenue variability.

Overall Sales Performance and Transaction Behavior

The store generated a total revenue of ₹12,19,396 over the study period from 2,378 sales transactions, an average of ₹512.78 per transaction. Monthly revenue (Figure 1) varies considerably rather than following a steady pattern. Distinct spikes appear in specific months, reflecting the influence of bulk wholesale transactions and seasonal demand, which fits the dual B2B and B2C nature of the business.

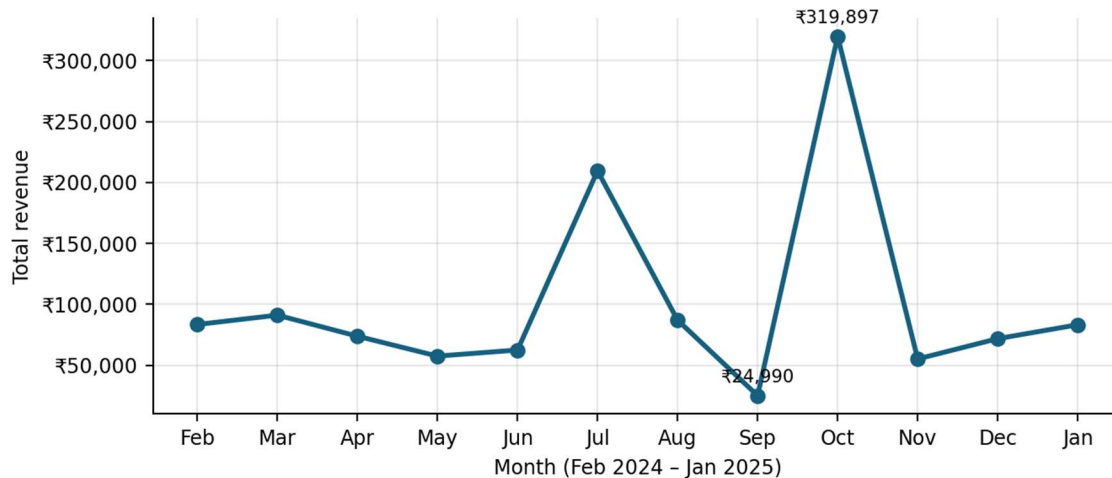


Figure 1: Trend of Total Revenue (per month) Across the Study Period

The highest monthly revenue, ₹3,19,897, was recorded in October, a significant spike compared with surrounding months. Lower revenue is observed in months such as May and June, reflecting subdued activity, and September is the lowest month at ₹24,990. These fluctuations show that total revenue depends strongly on a limited number of high-value periods and wholesale purchases rather than on consistent daily sales alone.

The histogram of transaction revenue (Figure 2) is strongly right-skewed. Most transactions (1,325 of 2,378, or 56%) fall in the ₹0–₹499 range, indicating that most sales are low-value purchases. As transaction value increases, frequency declines sharply, with only 7 transactions of ₹2,500 or more. The maximum transaction value is ₹3,250, reflecting occasional bulk wholesale purchases.

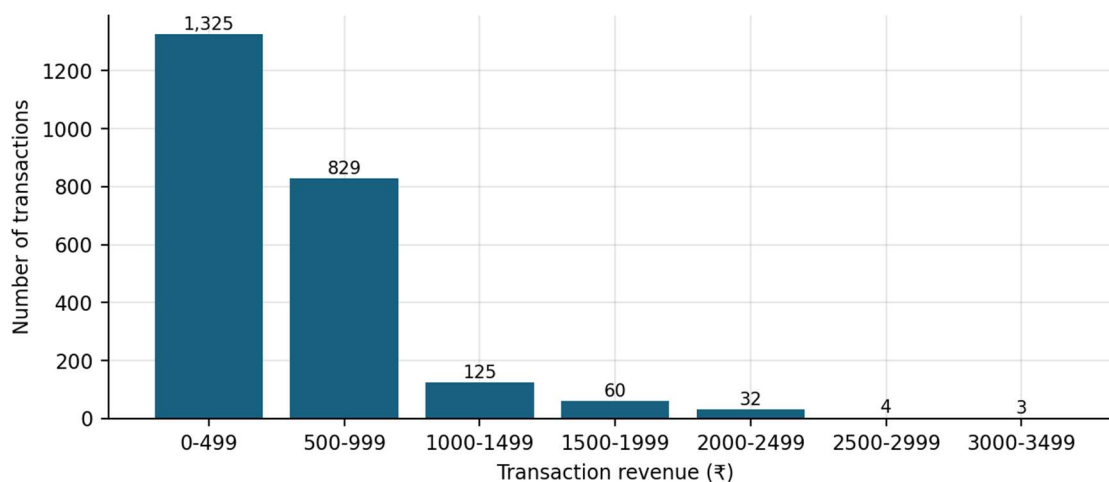


Figure 2: Distribution of Transaction-Level Revenue

Product Category-wise Sales Performance

Bangles contribute a significantly larger share of revenue than wool. Of the total ₹12,19,396, bangles account for ₹8,97,620 (73.6%) and wool products for ₹3,21,776 (26.4%). Bangles are the primary revenue-generating category, driven by higher-value transactions and bulk wholesale purchases.

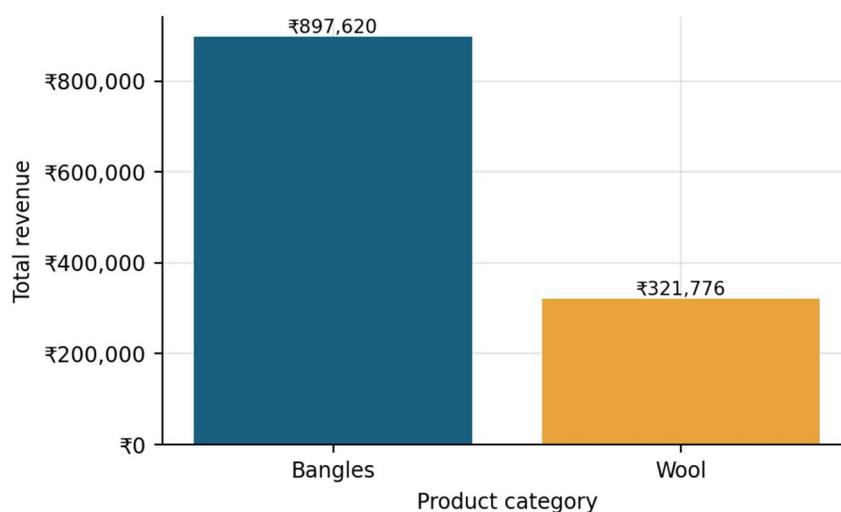


Figure 3: Product Category-wise Revenue Contribution

Figure 4 shows how revenue from each category changes across seasons. Bangles earn strong revenue in the Autumn/Festive season (October) and the Monsoon season (July–September), reflecting festival demand and wholesale buying. Bangle sales are nevertheless uneven across the year: October and July together account for 56% of bangle revenue, while November to February averages only about ₹20,355 a month.

Wool has a much stronger seasonal pattern. Sales stay low in Autumn, Monsoon and Summer but rise sharply in Winter (November–February), when wool makes up 72% of total revenue. Wool sales therefore depend heavily on cold weather.

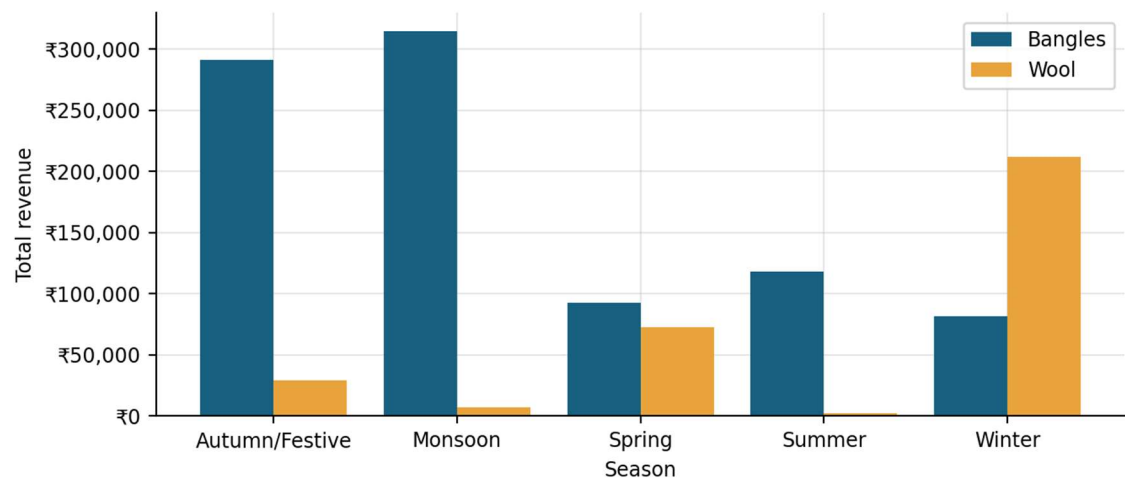


Figure 4: Seasonal Distribution of Revenue by Product Category

Overall, the two categories complement each other: bangles generate the large festive and wholesale peaks, while wool fills the winter gap when bangle sales are low (about ₹53,000 a month of wool revenue in winter), helping the business sustain revenue across the year.

Quantity Sold and Volume-driven Trends

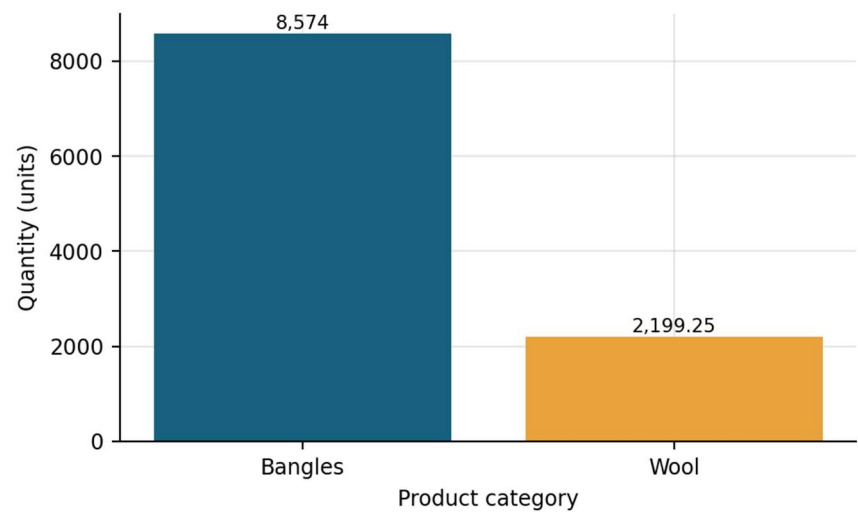


Figure 5: Product Category-wise Quantity Sold

Figure 5 presents the quantity sold for each category. Bangles account for a substantially higher quantity, 8,574 units (boxes), compared with 2,199.25 units of wool products (yarn, packets, jars and pieces). This reflects the nature of bangle sales, which are often conducted in bulk, particularly through wholesale transactions. Wool products are sold in smaller, need-based purchases, especially in colder

months. Units differ across products (boxes, yarn, pieces), so these quantities should be read as indicative; revenue is the like-for-like comparison. The contrast is important for inventory handling and storage planning.

Seasonal and Temporal Patterns

Seasonality plays a critical role in shaping sales at Naina Kangan and Wool Store. Time-based analysis at monthly and seasonal levels highlights clear demand cycles.

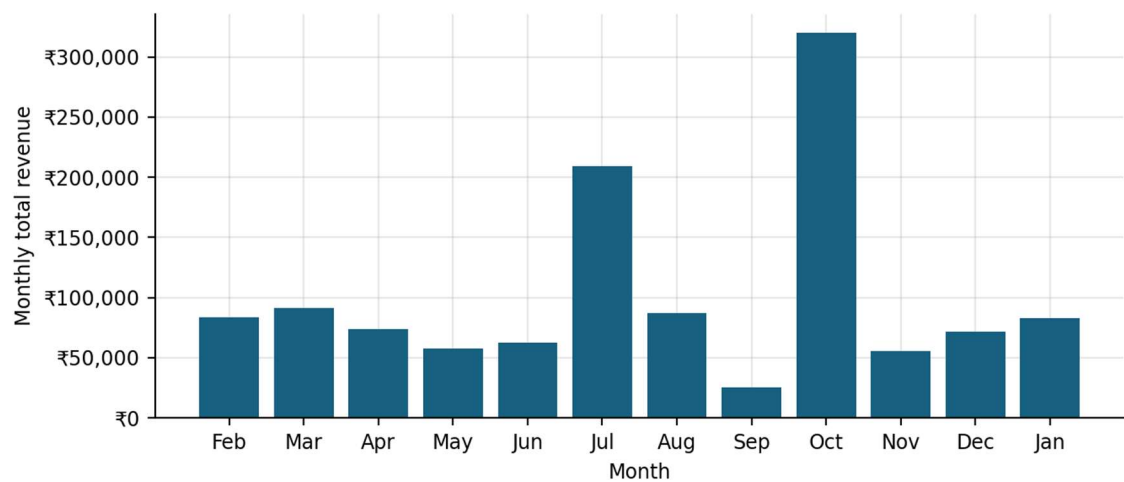


Figure 6: Monthly Revenue Trend

Figure 6 depicts the variation in total revenue by month. October records the highest revenue at ₹3,19,897, aligning with the festive season when wholesale bangle purchases increase significantly. September records the lowest at ₹24,990, a lean period before the festive surge. July (₹2,09,570) is the second-largest month, also bangle-driven. Wool-driven gains appear in the winter months (November to February).

To consolidate these observations, seasonal aggregation was performed (Figure 7).

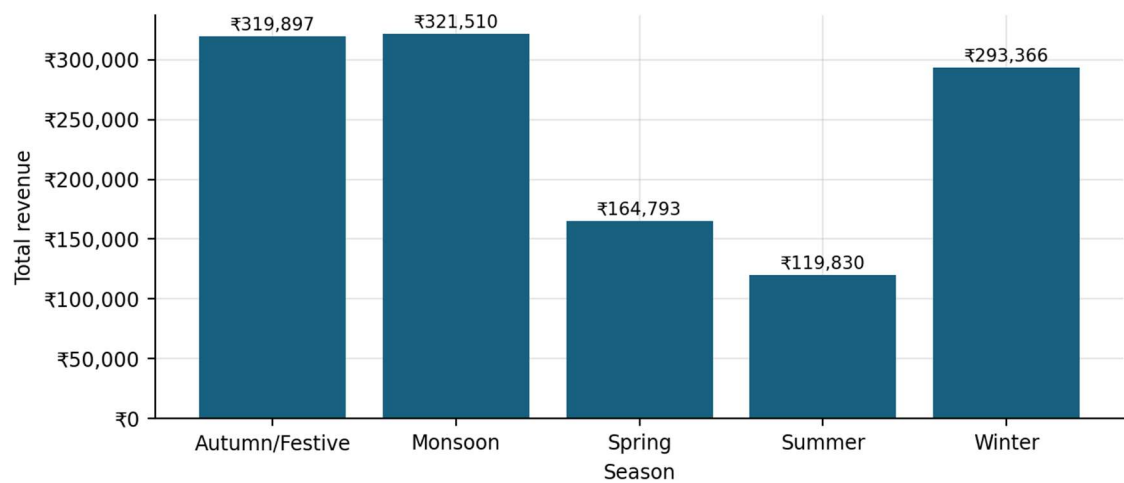


Figure 7: Season-wise Revenue Distribution

Monsoon (₹3,21,510 over three months) and Autumn/Festive (₹3,19,897 in October alone) are the highest-revenue seasons. On a per-month basis, Autumn/Festive is by far the strongest (about ₹3.2 lakh a month, against ₹1.07 lakh for Monsoon and ₹73,342 for Winter), with a higher average revenue per transaction (₹700) and greater transaction variability. This confirms that performance is closely tied to seasonal demand rather than evenly distributed through the year.

Customer Type Analysis (B2B vs B2C)

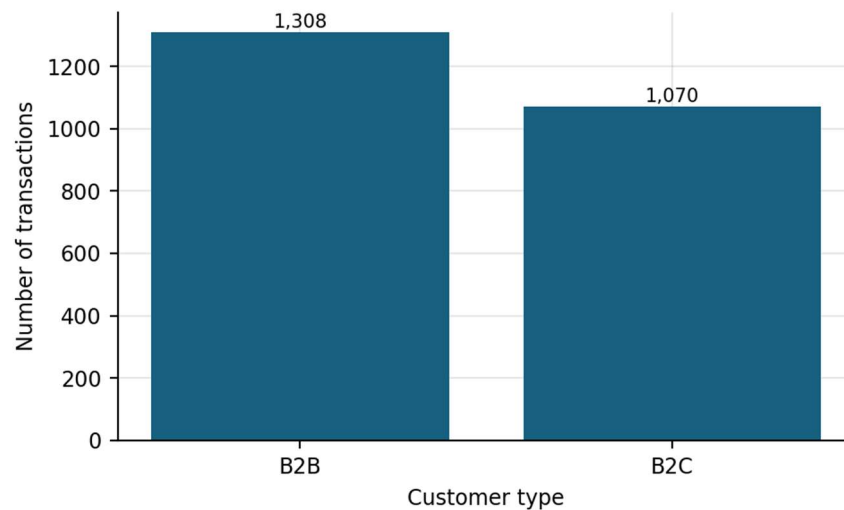


Figure 8: Transaction Frequency by Customer Type

B2B customers account for more transactions, 1,308, than B2C customers, 1,070. Business customers therefore transact more often; whether this reflects repeat buying cannot be verified because customer identifiers were not recorded.

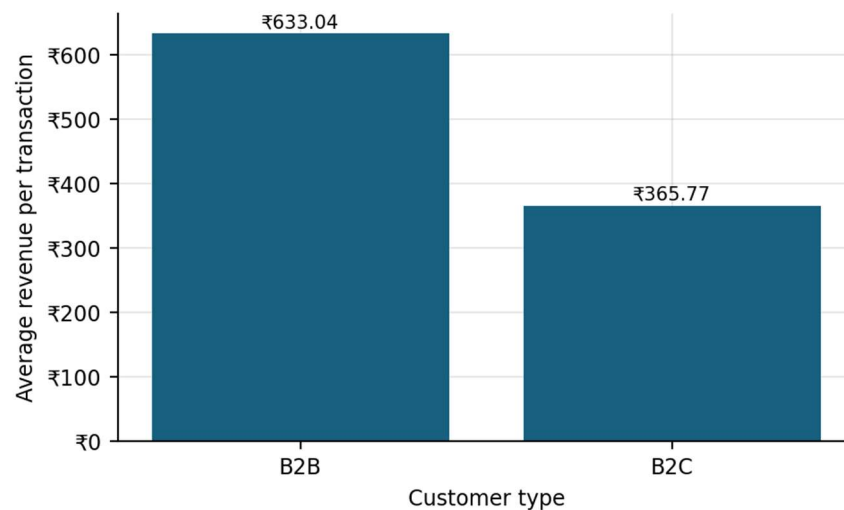


Figure 9: Average Revenue per Transaction by Customer Type

B2B customers contribute a significantly higher average revenue per transaction, ₹633.04, against ₹365.77 for B2C. B2B customers lead on both frequency and value and account for 68% of total revenue.

Product Type-wise Contribution to Gross Profit

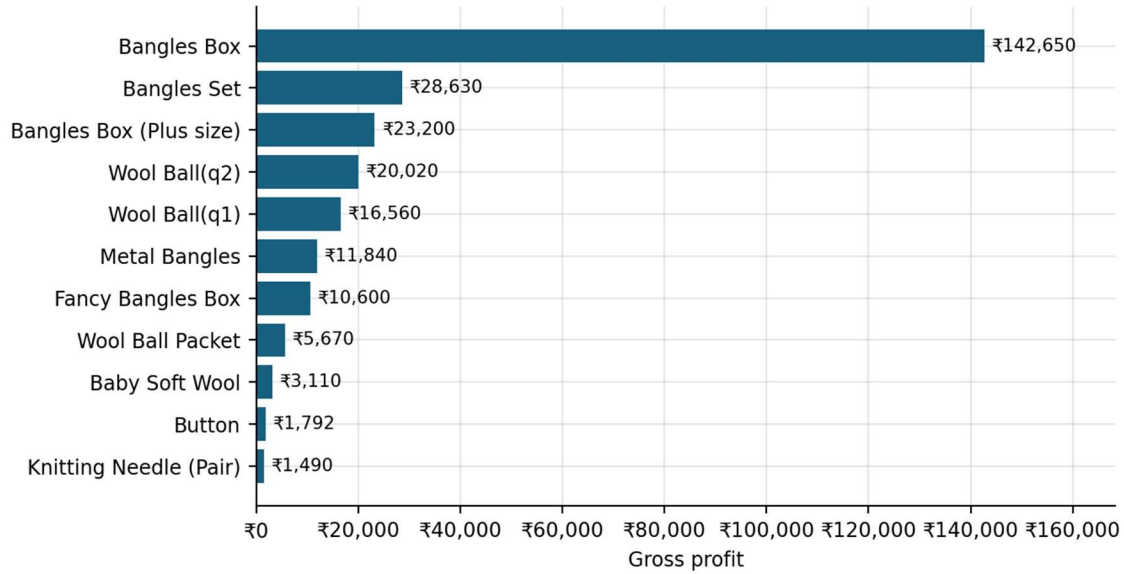


Figure 10: Gross Profit by Product Type

Figure 10 shows gross profit by product type. Bangles Box is the most profitable product, with ₹1,42,650 (53.7% of gross profit, from 50.7% of revenue). Knitting Needle (Pair) contributes the least, ₹1,490. Profitability is therefore not evenly distributed: a few bangle variants contribute a disproportionately high share of gross profit. Wool-based product types contribute smaller shares individually but provide stability collectively, particularly in winter.

Payment Characteristics and Transaction Completion

Share of transactions by payment mode

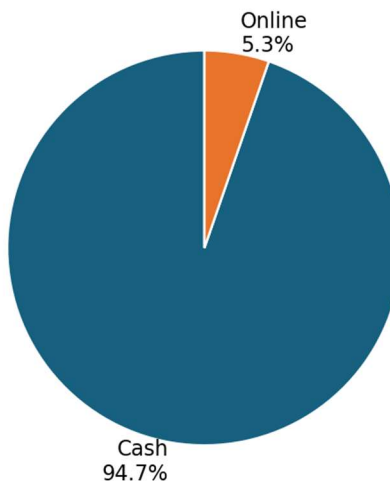


Figure 11: Payment Mode Distribution

Cash dominates, accounting for about 95% of transactions (2,252 of 2,378; 96% of revenue), while online payments account for the remaining 5% (126 transactions). This shows a strong customer preference for cash and limited adoption of digital payment.

Revenue Concentration and Dependence Patterns

An additional analysis examined how concentrated revenue is across transactions. The Pareto curve (Figure 12) shows moderate concentration: the top 10% of transactions contribute 28.8% of revenue and the top 20% contribute 44.6%, and about 54% of transactions are needed to reach 80% of revenue. Transactions of ₹1,000 or more (224 transactions, 9.4% of the total) contribute 27.7% of revenue. Concentration is stronger at product level (Bangles Box alone is 50.7% of revenue) and at seasonal level (October and July together are 43% of revenue).

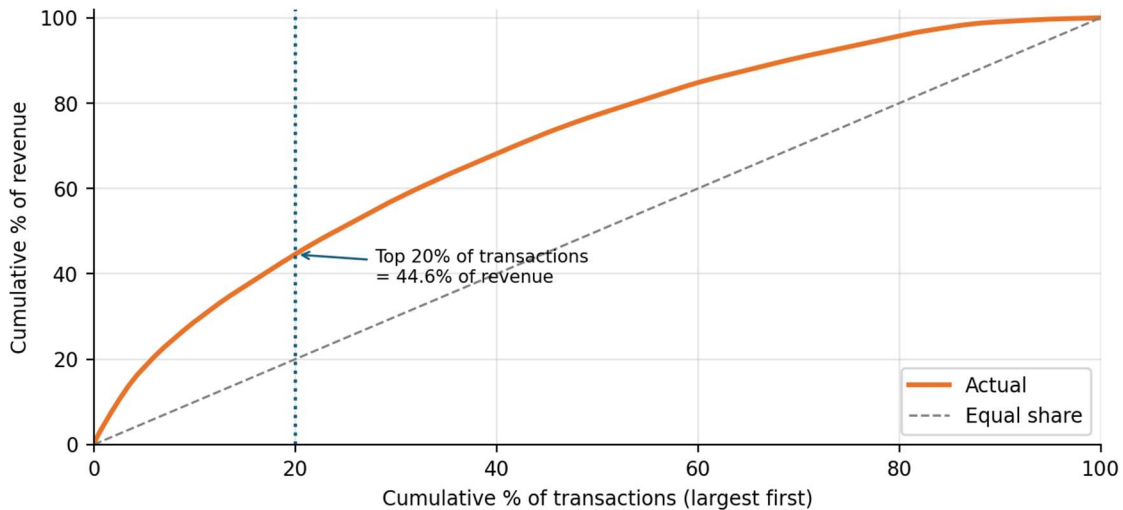


Figure 12: Pareto Curve of Transaction-wise Revenue Distribution

Integrated Findings and Business Implications

The combined analysis across time, product categories and customer types shows a mixed sales structure. Wool ensures strong winter performance, bangles support high-value wholesale revenue and the festive peaks, and retail customers provide steady smaller transactions. The business is fairly balanced across the year but strongly influenced by seasonal demand.

These results indicate a need for better inventory planning, season-based purchasing and different strategies for wholesale and retail customers. They provide a practical base for the next section, where the analytical tools are used to address inventory mismatch, changing demand and dependence on a few major revenue sources.

4. Interpretation of Results and Recommendation

Interpretation and Recommendations for Seasonal Sales Fluctuation

The analysis clearly shows that the business is highly seasonal. Total annual revenue of ₹12,19,396 is not evenly distributed: October (₹3,19,897) and July (₹2,09,570) together account for 43% of the year, while September records the lowest revenue at ₹24,990. This contrast between peak and off-peak periods confirms that seasonal demand is the dominant driver of sales performance.

Wool lifts winter revenue from a bangle baseline of about ₹20,000 a month to ₹55,000–₹83,000 a month, whereas the largest peaks (October and July) are bangle- and wholesale-driven. Without structured planning, stock and purchasing decisions rest on assumptions rather than historical demand, increasing the likelihood of overstocking wool after winter and understocking bangles ahead of festivals, with missed sales and inefficient use of working capital.

Season / Month	Gross Profit (₹)
Autumn/Festive	73,554.00
Oct	73,554.00
Monsoon	77,837.00
Jul	51,285.00
Aug	20,973.00
Sep	5,579.00
Spring	32,874.00
Mar	16,574.00
Apr	16,300.00
Summer	29,444.00
May	13,972.00
Jun	15,472.00
Winter	51,853.00
Jan	14,553.00
Feb	14,990.00
Nov	10,227.50
Dec	12,082.50
Grand Total	2,65,562.00

Table 1: Month-wise Gross Profit with Seasonal Classification

Recommendation: Adopt basic seasonal demand forecasting using historical monthly sales data. Plan winter wool procurement in advance from past revenue and quantity trends, plan bangle stock ahead of July and October, and reduce procurement in lean months. Even a simple moving average, year-on-year comparison or the monthly seasonal index will improve availability in peak months and reduce excess inventory in lean ones, while aligning purchases with expected cash flow. Because only one year of data is available, the index should be updated once a second year has been recorded.

Interpretation and Recommendations for Unsystematic Stock Handling

Product-wise analysis shows a clear imbalance in contribution. Bangles account for ₹8,97,620 of revenue against ₹3,21,776 for wool, and bangles also sold the larger quantity (8,574 units against 2,199 for wool). Bangles are therefore both higher-volume and higher-value, driven mainly by wholesale transactions, with a gross margin of 24.2% against 15.1% for wool, which is lower-volume, lower-margin and strongly seasonal. The business currently treats stock procurement uniformly across product types, which does not reflect these differing dynamics.

Product Type	Revenue (₹)	Quantity	Gross Profit (₹)
Baby Soft Wool	15,550.00	77.75	3,110.00
Bangles Box	6,18,150.00	4,755	1,42,650.00
Bangles Box (Plus size)	69,600.00	464	23,200.00
Bangles Set	1,02,250.00	409	28,630.00
Button	5,376.00	896	1,792.00
Fancy Bangles Box	79,500.00	2,650	10,600.00
Knitting Needle (Pair)	4,470.00	149	1,490.00
Metal Bangles	28,120.00	296	11,840.00
Wool Ball Packet	48,600.00	162	5,670.00
Wool Ball(q1)	1,07,640.00	414	16,560.00
Wool Ball(q2)	1,40,140.00	500.5	20,020.00
Grand Total	12,19,396.00	10,773.25	2,65,562.00

Table 2: Product-wise Quantity, Revenue, and Gross Profit Contribution

To make stocking decisions explicit, product types were classified by ABC-XYZ analysis (Table 3). Four products (Bangles Box, Wool Ball (q2), Wool Ball (q1) and Bangles Set) form Class A and generate 79% of revenue. Two products are Class B and five are Class C. No product has stable (X) demand: every product's monthly sales swing widely with the season, and several sit close to the 1.0 boundary between Y and Z, so Y and Z should be read as a spectrum rather than a hard line.

Product Type	Revenue share	Cumulative share	ABC	CV	XYZ	Class
Bangles Box	50.7%	50.7%	A	1.08	Z	AZ
Wool Ball(q2)	11.5%	62.2%	A	1.02	Z	AZ
Wool Ball(q1)	8.8%	71.0%	A	0.98	Y	AY
Bangles Set	8.4%	79.4%	A	1.54	Z	AZ
Fancy Bangles Box	6.5%	85.9%	B	1.30	Z	BZ
Bangles Box (Plus size)	5.7%	91.6%	B	1.11	Z	BZ
Wool Ball Packet	4.0%	95.6%	C	0.96	Y	CY
Metal Bangles	2.3%	97.9%	C	1.70	Z	CZ
Baby Soft Wool	1.3%	99.2%	C	0.94	Y	CY
Button	0.4%	99.6%	C	1.34	Z	CZ
Knitting Needle (Pair)	0.4%	100.0%	C	0.99	Y	CY

Table 3: ABC-XYZ Classification of Product Types (monthly units, Feb 2024 – Jan 2025)

Class A items with highly seasonal demand (AZ) should be stocked ahead of their peak months and pre-booked with wholesalers rather than held all year. Class B items should be ordered in smaller lots ahead of season and replenished as they sell. Class C items (together about 8% of revenue) should be ordered on demand to avoid tying up capital. Stock-outs of Bangles Box would hurt most, since it alone contributes 50.7% of revenue.

Recommendation: Adopt a differentiated stock-planning approach based on ABC-XYZ class. Prioritize Class A products in inventory decisions, especially in periods of known wholesale demand; stock wool in line with seasonal demand rather than uniformly through the year; and review product-wise contribution regularly to identify slow-moving items early.

Interpretation and Recommendations for Limited Customer Acquisition Channels

Customer-type analysis shows that B2B customers account for 1,308 transactions and B2C customers for 1,070. Wholesale customers lead on both transaction count and value per transaction (₹633.04 against ₹365.77 for retail) and contribute 68% of revenue. However, the absence of customer-level records prevents the business from identifying repeat wholesale buyers or analyzing retention. Dependence on word of mouth alone restricts growth and increases vulnerability to demand uncertainty.

Without knowing which customer segment contributes more to long-term profitability, decisions on pricing, inventory prioritization and customer engagement remain intuitive rather than data-driven.

Customer Type	Transactions	Average Revenue (₹)	Total Revenue (₹)
B2B	1,308	633.04	8,28,020
B2C	1,070	365.77	3,91,376
Grand Total	2,378	512.78	12,19,396

Table 4: Customer Type vs Transaction Count and Average Revenue

Recommendation: Begin maintaining basic customer records, particularly for wholesale buyers: customer type, purchase frequency and average transaction value. Strengthening relationships with repeat wholesale customers can improve revenue stability, while understanding retail purchase patterns can help optimize pricing and availability. WhatsApp Business and a Google Business profile are low-cost ways to reach new retail and wholesale customers. Even minimal customer segmentation will improve decision-making and long-term growth prospects, and will make RFM analysis possible in future.

Interpretation of Payment and Transaction Characteristics

About 95% of transactions are cash-based and only 5% use online payment. This reflects the traditional nature of the business, where immediate payment at the point of sale is preferred. While this reduces default risk, it limits traceability and opportunities for digital customer engagement. Cash dominance indicates strong liquidity but also dependency on manual record-keeping, which raises the chance of recording errors and reduces the scope for automated analysis or integration with digital accounting systems.

Recommendation: Gradually adopt digital payment modes to improve transaction tracking and financial transparency without disrupting customer preferences. Encouraging digital payment for wholesale transactions in particular will improve record accuracy and support future integration with accounting and inventory tools.

Operational Efficiency and Decision-Making Maturity

The analysis reveals a gap between operational intensity and decision-making maturity. The business handles a considerable number of transactions and maintains stable relationships with retail and wholesale customers, yet most decisions rely on experience and intuition rather than structured inputs. While this experience is valuable, the lack of simple planning tools places a high decision burden on the owner and limits systematic growth.

Even basic descriptive analysis, such as average transaction values, monthly revenue patterns and product-wise contribution, reveals useful and actionable insights. Improving decision-making does not require complex software: small, gradual use of data for planning can significantly improve efficiency while remaining realistic for a small retail business. Converting routine sales records into structured summaries supports decisions such as adjusting stock purchases ahead of peak demand, recognizing slow periods early, and focusing on the products and customer segments that contribute most to revenue.

Integrated Interpretation and Business Implications

The combined interpretation of sales trends, product performance and customer behavior shows a mixed but highly seasonal structure. Wool provides strong winter demand, bangles contribute high-value wholesale revenue and festive peaks, and retail customers provide steady smaller transactions. Reliance on intuition-based decision-making exposes the business to inventory inefficiencies, product and seasonal revenue concentration, and missed growth opportunities.

By aligning stock planning with seasonal demand, prioritizing high-revenue products through ABC-XYZ classification, and strengthening customer record-keeping, the business can move toward structured, informed decision-making. The business has strong revenue-generating capability but needs improved analytical structure to sustain and optimize performance. Integrating seasonal planning, product prioritization and customer-focused strategies can transform reactive decision-making into proactive management and support long-term sustainability.

Limitations

- Only one year of data is available, so seasonal patterns describe this year and are not yet a validated forecast.
- No customer identifiers were recorded, so repeat-customer behavior and RFM segmentation could not be analyzed.
- Units differ across products (boxes, yarn, packets, pieces), so quantities are not comparable across categories; revenue is.
- Records were digitized by hand from handwritten registers.